Differential Power Processing for Efficient Data Center Power Conversion

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Differential Power Processing

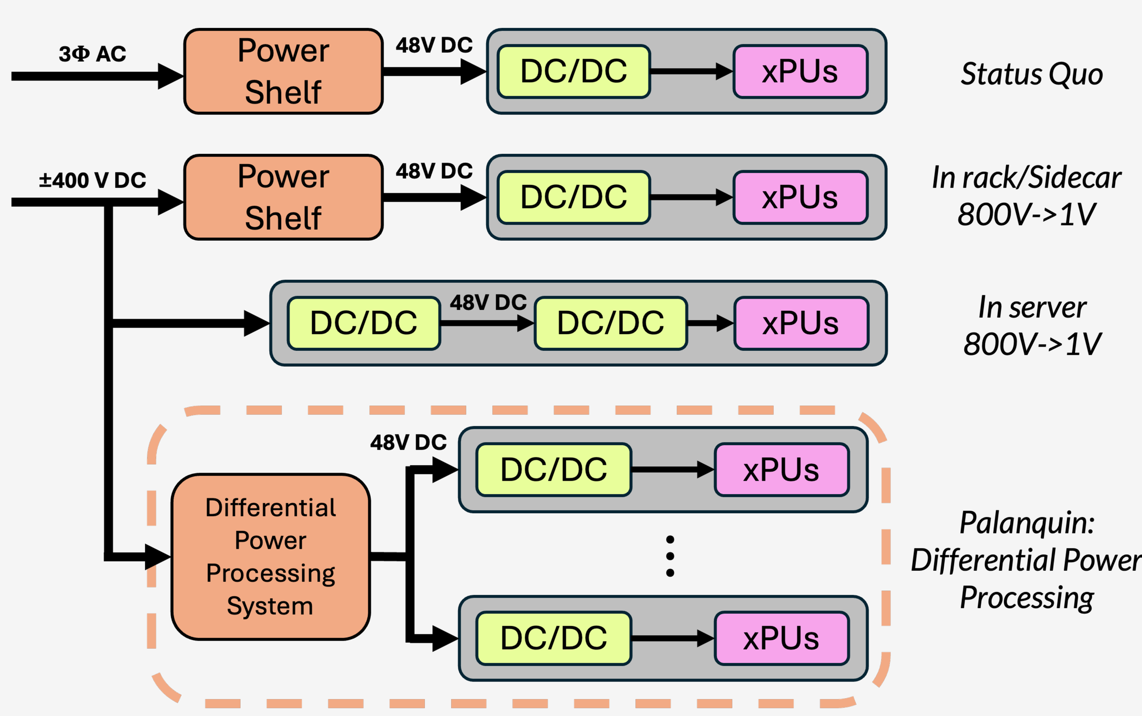

Figure 1. Comparison of status quo and potential future rack-level power distribution architectures.

Palanquin Power is developing a novel rack-level power delivery system, capable of achieving **99+% efficiency at high power densities**. We utilize a differential power processing (DPP) architecture, described below. This approach is unique in that it decouples system efficiency from converter efficiency.

In our system, servers are connected in series and current is fed directly to the stack from the rack level bus. Power can thus be delivered to loads with no losses from intermediate converters. Each load interfaces with an isolated, bidirectional converter that regulates its voltage. Critically, converters only process the difference in power between servers. This means system efficiency can be extremely high in cases where loads are well-matched, such as AI training and other high utilization tasks.

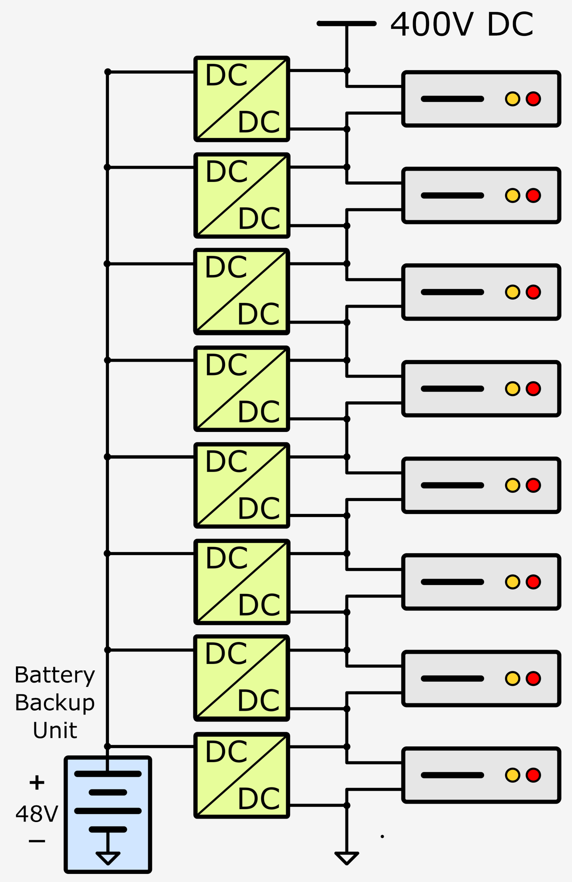

Figure 2. Diagram of a differential power processing system for server rack applications. Note that individual PSUs must be isolated & bidirectional.

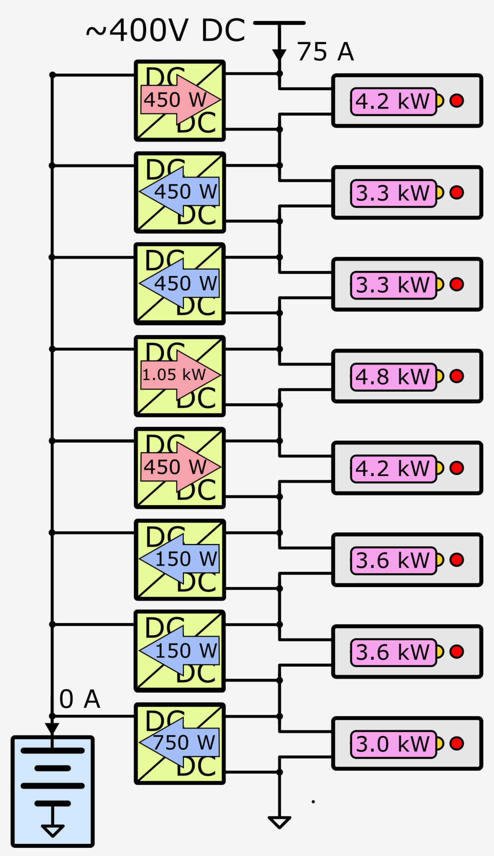

Figure 3. Diagram showing power flow in a DPP system. Note that the power processed by converters is significantly smaller than the power delivered to the servers.

Motivation

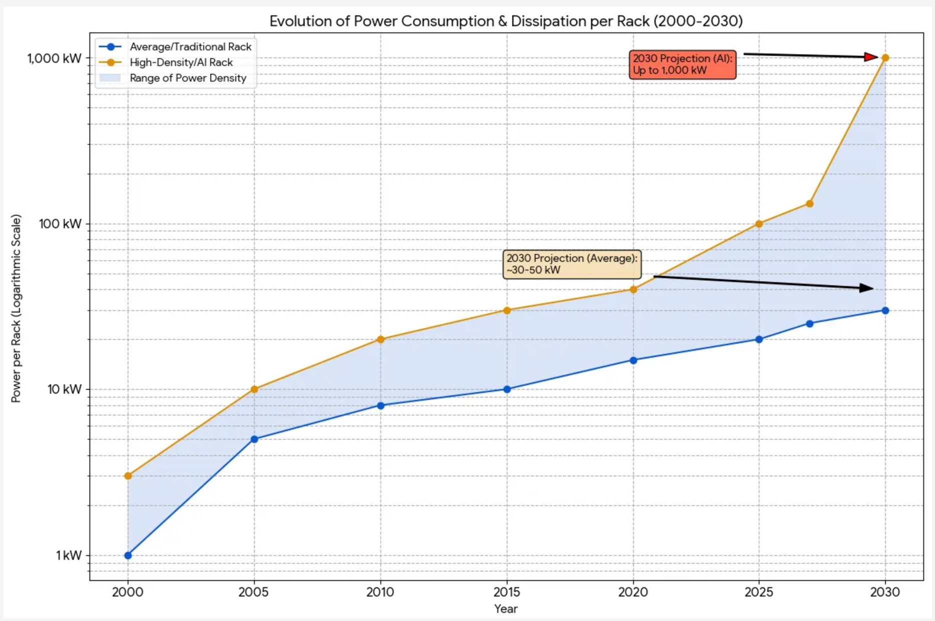

Figure 4. Power consumption of an AI and conventional server racks over time.

- Rack power density is rapidly increasing from ∼100 kW/rack to 1 MW/rack, driven by the needs of AI CPUs/GPUs. This necessitates advancements to power architecture design.
- Efficiency is highly valuable, given limited power availability for data centers. However, efficiency is typically traded off against power density in converter design.
- AI applications also demand high peak transient power, increasing the burden on power supply units (PSUs).
- Significant changes to power architecture - namely ± 400V or 800V DC distribution enabled by solid state transformer technologies - are on the horizon, and it may be an appropriate time for integration of novel approaches.

Selected References

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Hardware Demonstration

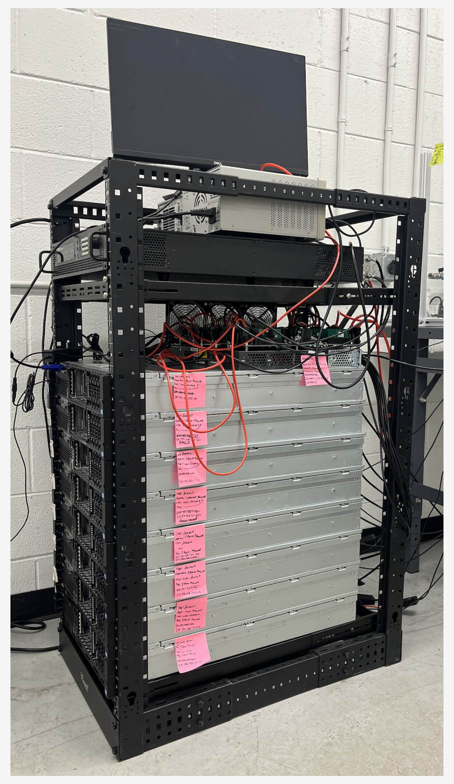

Figure 5. DPP lab demonstration delivering power to live servers.

We built a demonstration system using capable of delivering well-regulated power to real servers at high power & high efficiency. Fig. 5 shows the Palanquin power delivery system configured with a DC power supply configured as a 200V DC input; a DC power supply and electronic load acting as a synthetic 50V battery energy storage unit; and eight servers with 50V DC inputs configured in four series connected voltage domains.

Figs. 7 & 8 show system voltage and current waveforms during turn-on and a transient event with server loads, ramping up to ∼2 kW of total load. Voltages remain stable & well-regulated. Note furthermore that system control is completely decentralized.

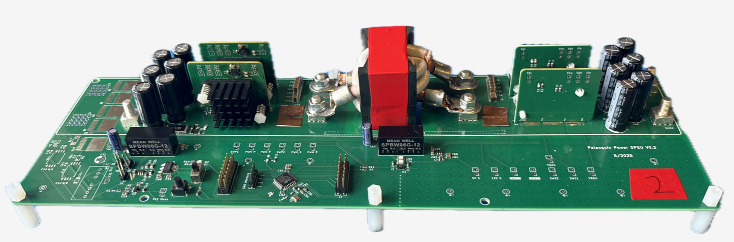

Figure 6. 3 kW bidirectional, isolated DC/DC unit DPP converter prototype.

In an alternative setup using DC electronic laods, the system achieves peak efficiency of 98.5%, corresponding to about 125W of total loss. This occurs despite utilizing power converters with ∼96% peak efficiency.

Losses are dominated by steady state (gate driving, ancillary power) and conduction losses. We anticipate that with a combination of A) a converter optimized for light load operation; B) optimized conduction paths; C) even higher power operation, 99+% efficiency could be achieved.

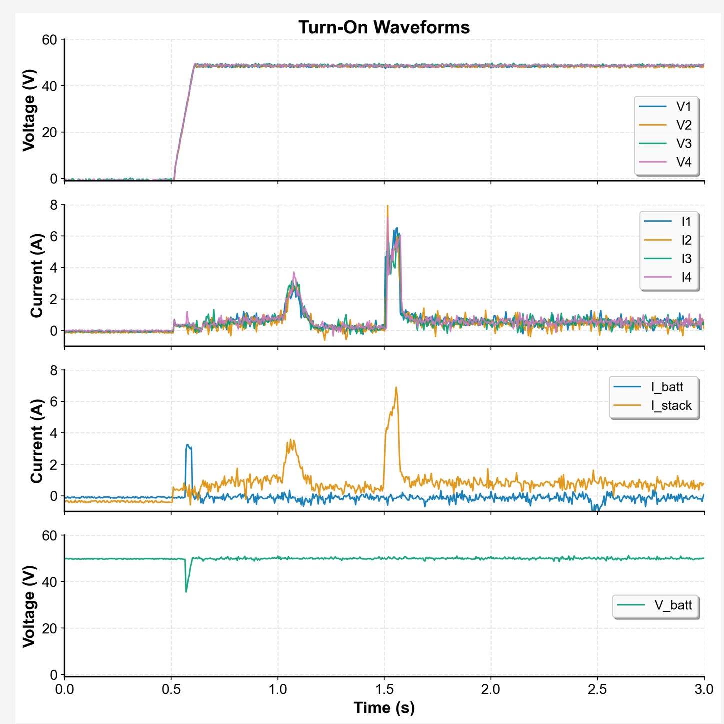

Figure 7. Experimental data from system turn-on, showing stable voltage regulation.

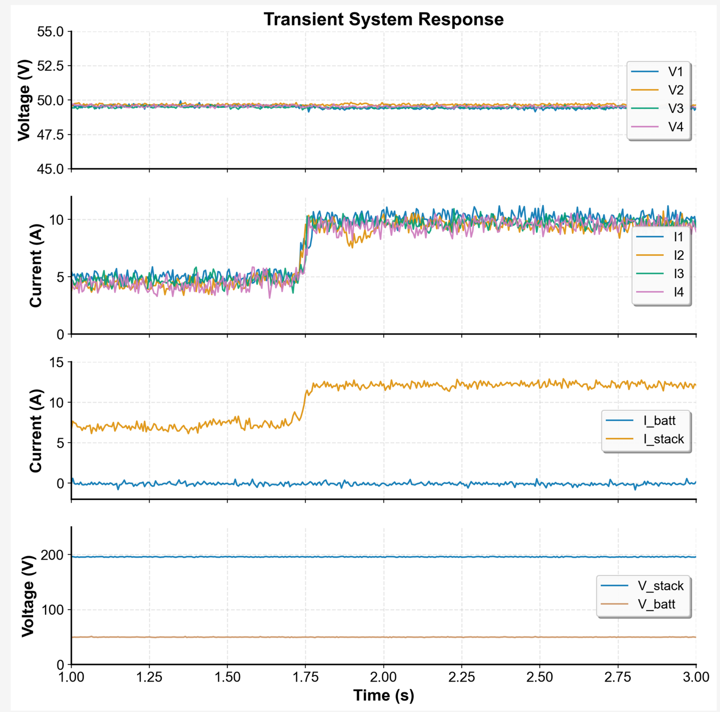

Figure 8. Waveforms showing stable voltage regulation during transient event on operational servers.